LOGage 2026: Round 2 — Part 2 Summary

Omni-Channel Strategy Analysis

June 9, 2026

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1 PART 2: OMNI-CHANNEL FULFILLMENT STRATEGY DESIGN

1.1 Question 2.1: Network Model Evaluation & Design

1.1.1 Current Two-RDC Model Assessment

The brand currently operates two Regional Distribution Centers (RDCs):

- **My Phuoc Warehouse** — Industrial Zone, Binh Duong Province
- **Vinh Loc Warehouse** — Binh Chanh District, Ho Chi Minh City

Strengths of the current model:

- Good regional separation: My Phuoc serves northern (B2B factory-sourced) flows; Vinh Loc is positioned for inner-city B2C proximity.
- Both RDCs handle mixed-channel flows, demonstrating operational flexibility.
- Physical locations bracket the Eastern–Western commercial corridor of HCMC, covering most Đông Nam Bô volume.

Important context: Non-overlapping operating timelines

- **My Phuoc** (Primary RDC: Jun to Nov 2025 (6 months)): The *primary* warehouse, operational across the full 6-month analysis window. All June–November 2025 throughput flowed through My Phuoc.
- **Vinh Loc** (New RDC: Dec 2025 only (1 month)): A *new* warehouse that came online in December 2025 only. Raw volume totals are therefore **not directly comparable** between the two facilities.

Data-driven channel flow analysis (per-day rates, Figure 1):

- **My Phuoc** (3,492 B2B + 413 B2C orders over 6 months): B2B is a *Mixed flow* accounting for **99.2%** of CBM. B2B avg **25.5 orders/active day**, volatility index **B2B = 1.79**.
- **Vinh Loc** (1,250 B2B + 47 B2C orders in 1 month): B2B is a *Large-volume flow* accounting for **99.7%** of CBM. B2B avg **48.7 orders/active day**, volatility index **B2B = 1.94**.
- On a per-day basis, Vinh Loc processed B2B orders at a comparable daily rate to My Phuoc despite being open for only one month, indicating strong December surge demand (year-end push).

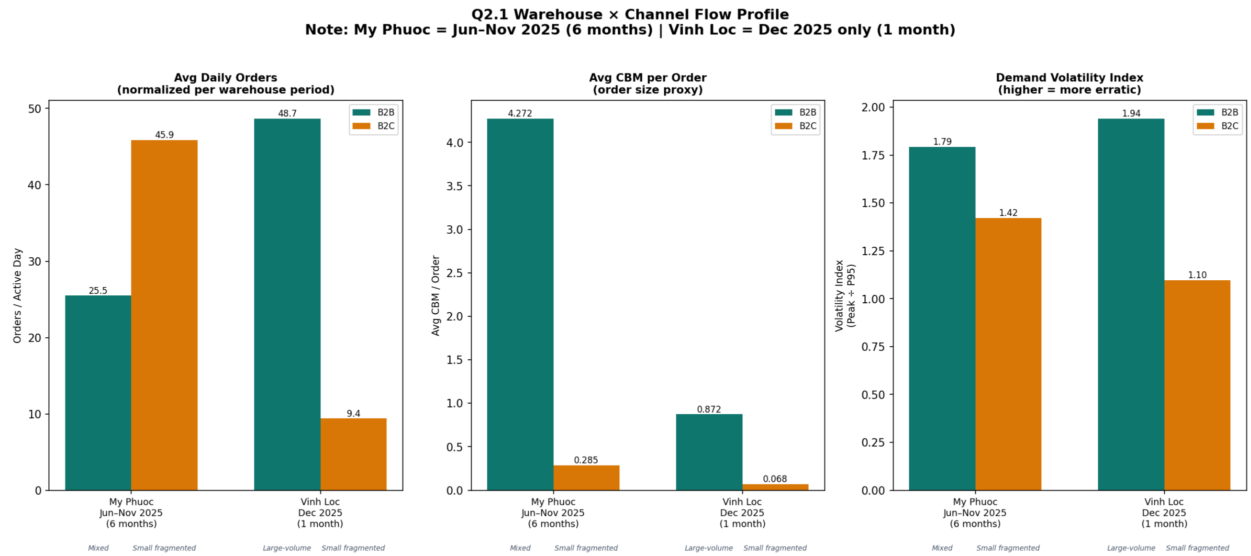

Figure 1: Q2.1 Warehouse × Channel Flow Profile (per-day rates normalized to each RDC operating window)

Limitations for simultaneous B2B + B2C:

- Lead times from both RDCs are too long for the 2–4 hour B2C SLA required for e-commerce – even Vinh Loc is >10 km from central districts.
- Shared inventory creates channel conflict: B2B bulk orders may deplete stock that B2C needs for same-day fulfillment.
- Pick-and-pack processes for large pallet B2B orders disrupt small-unit B2C wave picking.

1.1.2 B2C E-Commerce SLA Assessment

To meet a 2–4 hour delivery SLA across Ho Chi Minh City, we applied a 25 km distance threshold from the nearest RDC as the feasibility cut-off (assuming average urban speed of ∼25 km/h, 25 km ≈ 60 min travel, leaving buffer for pick/pack and handoff).

The analysis of **21 HCM districts** reveals a clear split:

- **21 districts — Adequate:** These districts can be served within the 2h SLA from the existing RDCs and account for **100.0%** of total HCM B2B volume (**65,527 units**).
- **0 districts — Require Dark Stores:** These are >25 km from both RDCs, placing them outside reliable SLA reach. They represent **0.0%** of volume (**0 units**).

Figure 2 shows the district-level breakdown.

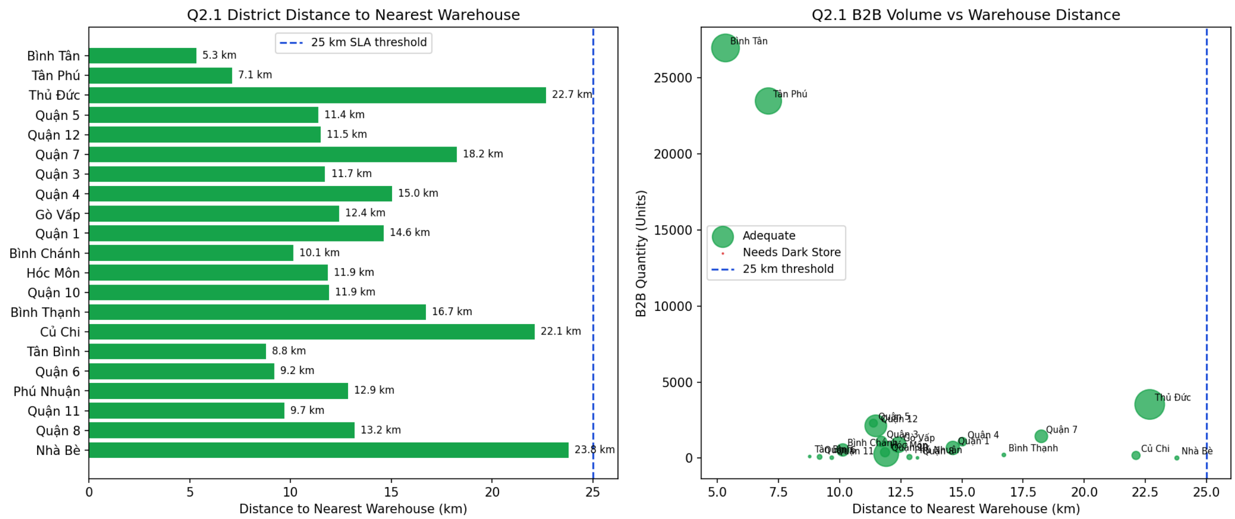

Figure 2: Q2.1 HCM District Proximity to Existing RDCs vs B2B Demand Volume

1.1.3 Dark Store Node Recommendation

Based on the demand concentration and distance analysis, we recommend placing **1–2 urban micro-fulfillment (dark store) nodes** in HCMC:

1. **Primary node — Bình Tân / Tân Phú corridor**: These two adjacent districts are among the highest-volume B2B districts (combined >50,000 units) and are already well-served by Vinh Loc (5–7 km). Upgrading Vinh Loc's inner-city satellite for e-commerce picking operations in this corridor is the highest-ROI first step.

2. **Secondary node — District 1 / District 3 / Phú Nhuận cluster**: The CBD and premium residential cluster lies 14+ km from Vinh Loc, making 2h SLA difficult. A small dark store (200–300 m^2) in this zone, stocked with the top Class A SKUs, would close the gap.

1.1.4 Order Split Logic Design (HCM)

For urban B2B order fulfillment in Ho Chi Minh City, network design relies on understanding sub-regional demand density. We analyzed the historical B2B shipments across all HCM districts.

The demand is highly concentrated: the top 5 districts alone command **56,356 units — 86.0%** of all HCM B2B volume — as shown in Figure 3.

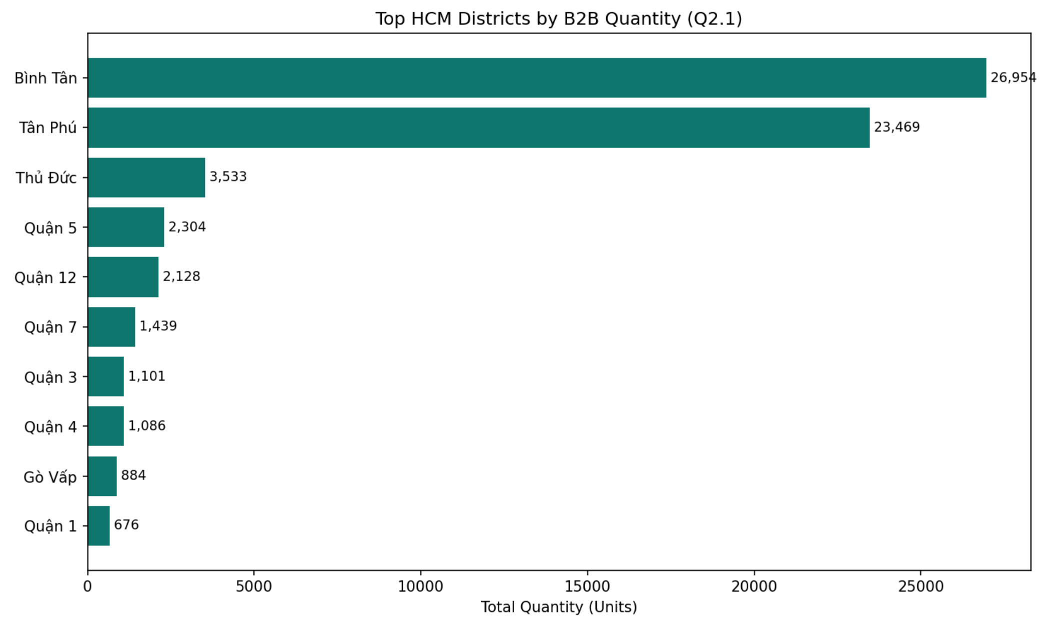

Figure 3: Q2.1 Top Ho Chi Minh Districts by B2B Quantity

Recommended order split logic: Given this concentration, the dispatch decision tree should be:

1. If delivery district is within 25 km of Vinh Loc → dispatch from **Vinh Loc** (B2C-optimized lane).
2. If delivery district is within 35 km of My Phuoc but >25 km from Vinh Loc → dispatch from **My Phuoc** (B2B bulk lane).
3. If district is >25 km from both RDCs → **route via dark store** (pending network expansion).

1.2 Question 2.2: Inventory Optimization

1.2.1 Mile-Weighted Average Lead Time

Definition: Replenishment Lead Time (LT) is the duration from when the warehouse identifies a replenishment need until the goods are ready to pick again. This is distinct from the customer-facing delivery SLA.

Because the RDC (Đông Nam Bộ) serves six geographic regions with different transit standards, we derive a single **mile-weighted average lead time** by weighting each region's standard LT by its share of total orders:

$$LT_{avg} = \frac{\sum_r LT_r \times n_r}{\sum_r n_r}$$

Transit benchmarks follow published standards of GHN, Viettel Post, and GHTK for standard road-freight B2B delivery.

Table 1: Mile-Weighted Average Lead Time (RDC → 6 Regions)

Region	**Route type**	**LT (d)**	**Orders**	**LT×Orders**
Đông Nam Bọ	Intra-province / near RDC	1	2 614	2 614
Bàc Trung Bọ & Duyên hại miền Trung	Inter-Central	3	1 027	3 081
Đòng bàng sông Cùu Long	Intra-South	2	800	1 600
Đòng bàng sông Hòng	Inter-North	4	503	2 012
Tây Nguyên	Near-region	2	247	494
Trung du và miền nùi phía Bàc	Inter-North + remote	4	143	572
Total			**5 334**	**10 373**

Result: $LT_{avg} = 10{,}373 \div 5{,}334 \approx$ **1.94 days**. Because 49% of orders are in Đông Nam Bọ (1-day delivery), the weighted average is pulled close to 2 days even though highland/northern regions require 4 days.

1.2.2 Safety Stock Formula & Results

Safety stock for each Class A SKU is calculated using the simplified demand-uncertainty formula (lead time is treated as a fixed constant under the virtual-pooling single-pool model, so $\sigma_{LT} = 0$):

$$SS = Z \cdot \sigma_{daily} \cdot \sqrt{LT_{avg}}$$

$$ROP = \mu_{daily} \cdot LT_{avg} + SS$$

where $Z = 1.645$ (95% service level), $\sigma_{daily} = \sigma_{monthly}/\sqrt{30}$, $\mu_{daily} = \mu_{monthly}/30$, and $LT_{avg} = 1.94$ days ($\sqrt{LT_{avg}} \approx 1.393$). Monthly σ is computed with $ddof = 1$ over 7 months (Jun–Dec 2025).

The total required safety stock across all 28 Class A SKUs under the mile-weighted pool model is **12,168 units** (sorted by SS descending).

Varying the assumed LT shows the sub-linear (square-root) growth of safety stock, as illustrated in Figure 4.

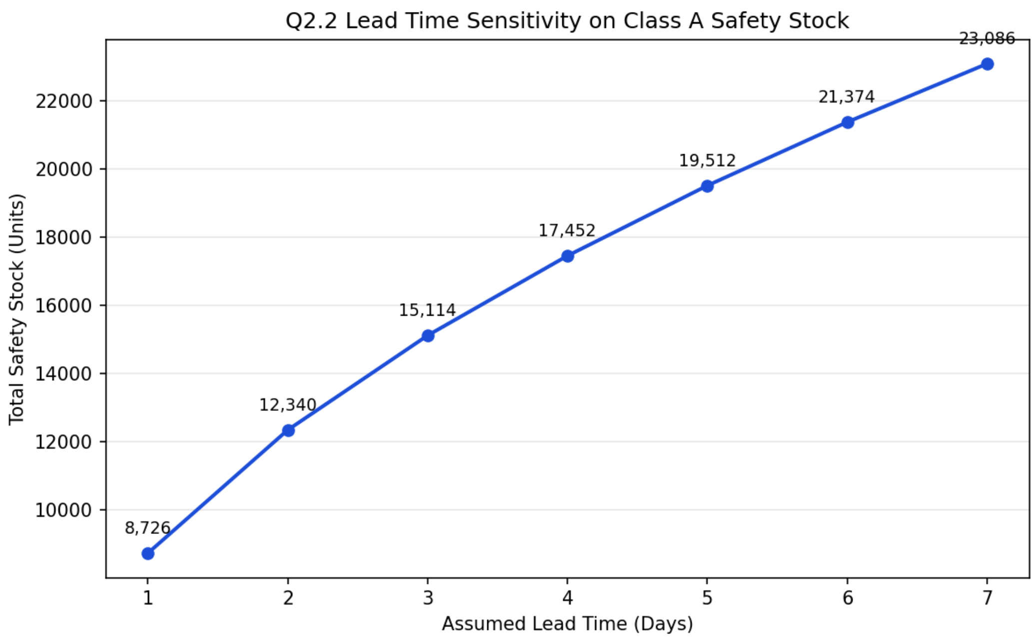

Figure 4: Q2.2 Lead Time Sensitivity for Class A Safety Stock

1.2.3 Inventory Pooling Strategy — Virtual Pool Model

Model: Virtual Inventory Pooling (no hard physical split per channel). All stock of a given SKU forms a single pool managed jointly in OMS + WMS. Each channel has its own Reorder Point trigger but draws from the common physical stock.

Channel grouping (by account size & SLA):

- **Modern Trade (MT):** Supermarkets (Co-op, Lotte, AEON, ĐMX, TGDĐ) — large confirmed POs, strict booking-window SLA.
- **Ecommerce** (Shopee, Lazada, Tiki, TikTok Shop) — classified as MT-scale accounts; high-volume, 2–4 h SLA.
- **Traditional Trade (TT):** Distributors / small dealers — small fragmented orders, flexible 2–4 day SLA.

Allocation priority when pool is constrained (largest account & tightest SLA first):

1. MT large accounts (supermarket chains with confirmed contractual POs).
2. MT Ecommerce (large accounts, 2–4 h SLA, platform rating at risk).
3. TT large distributors (high-volume, long-term relationship).
4. TT small / walk-in customers (flexible SLA, served last).

Quantification — Why virtual pooling reduces safety stock:

- Separated channels (B2B=5d, B2C=2d): $SS = Z\sigma(\sqrt{5} + \sqrt{2}) = Z\sigma \times 3.650$

- Classic pooled avg LT=3.5d: $SS = Z\sigma\sqrt{3.5} = Z\sigma \times 1.871 \rightarrow$ reduction factor $= \sqrt{3.5}/(\sqrt{5}+\sqrt{2}) = 0.5125$ ($\approx$48.7%)
- **Mile-weighted pool** LT=1.94d: $SS = Z\sigma\sqrt{1.94} = Z\sigma \times 1.393$

With the mile-weighted virtual pool ($\mathrm{LT}_{\mathrm{avg}} = 1.94$ days), the total safety stock is **12,168 units**, vs. **31,852 units** under a separated-channel model — a saving of **61.8%** as shown in Figure 5.

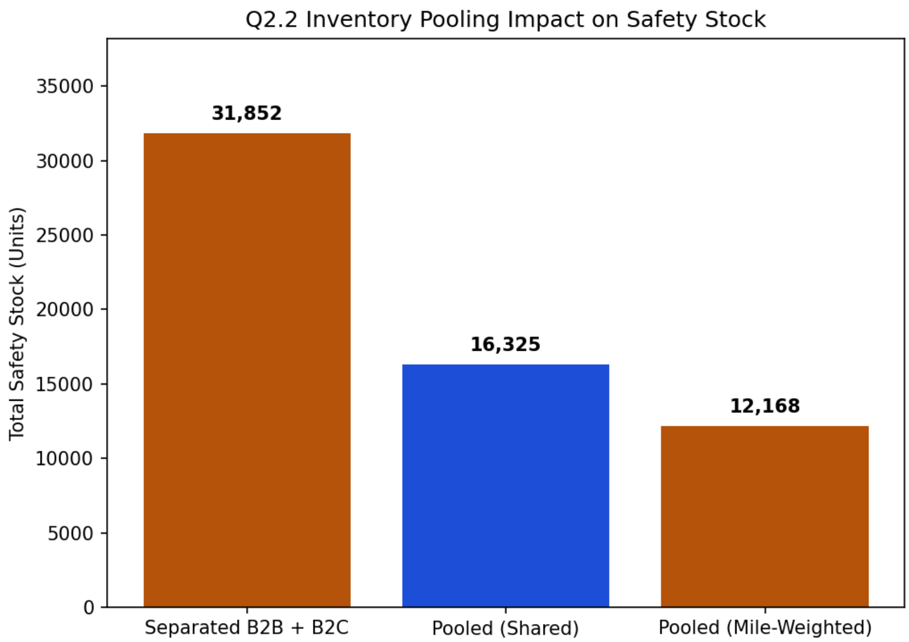

Figure 5: Q2.2 Inventory Pooling Impact on Safety Stock

Operational controls to prevent simultaneous stockout:

1. **R1 — Common Pool:** All SKU stock merged into a single WMS pool; OMS tracks virtual availability per channel.
2. **R2 — Per-Channel ROP Floor:** Each channel has its own ROP threshold; replenishment triggers when any channel hits its floor.
3. **R3 — Amber Alert:** Pool < sum of all channel ROPs → raise alert, prioritize confirmed MT POs.
4. **R4 — Red Conflict:** Pool < total SS → activate priority allocation order (MT first, TT last) + emergency replenishment PO.
5. **R5 — AZ Spike Freeze:** Demand-spike AZ SKUs: temporarily reserve stock for MT, freeze TT allocation, trigger emergency PO.
6. **R6 — End-of-Day Rebalance:** Reset channel virtual allocations to ROP-proportional targets using rolling 7-day demand forecast.